

Unit 4 Study Guide ITERATION 2

Linear Functions | Graphs, Standard Form, and Inequalities

Name: _____ Period: _____ Due: _____

How to use this guide: This iteration is aligned to the Unit 4 test: slope from points and equations, slope-intercept and point-slope form, vertical and horizontal lines, graphing equations, matching equations to graphs, and linear inequalities.

Multiple-Choice Practice

1A. Slope Through Two Points | Version A

Given the points (6, 4) and (7, 2), what is the slope?

- A. -2
- B. $-1/2$
- C. $1/2$
- D. 2

Work / thinking

1B. Slope Through Two Points | Version B

Given the points (-3, 5) and (1, -3), what is the slope?

- A. 2
- B. -2
- C. $1/2$
- D. $-1/2$

Work / thinking

2A. Slope from Standard Form | Version A

What is the slope of the line $3x - 4y = 16$?	Work / thinking
A. -4	_____
B. $\frac{3}{4}$	_____
C. $\frac{4}{3}$	_____
D. 3	

2B. Slope from Standard Form | Version B

What is the slope of the line $5x + 2y = 12$?	Work / thinking
A. $-\frac{5}{2}$	_____
B. $\frac{5}{2}$	_____
C. $-\frac{2}{5}$	_____
D. $\frac{2}{5}$	

3A. Slope-Intercept Form | Version A

Which expresses $4x + 2y = 8$ in slope-intercept form? A. $x = 1/2y + 2$ B. $x = -1/2y + 2$ C. $y = 1/2x + 2$ D. $y = -2x + 4$	Work / thinking _____ _____ _____
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3B. Slope-Intercept Form | Version B

Which expresses $6x - 3y = 9$ in slope-intercept form? A. $y = 2x - 3$ B. $y = -2x + 3$ C. $y = 6x - 9$ D. $y = -6x + 9$	Work / thinking _____ _____ _____
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4A. Equivalent Forms | Version A

Which equation is equivalent to $y - 4 = -\frac{1}{2}(x - 8)$? A. $y = -\frac{1}{2}x + 8$ B. $y = -\frac{1}{2}x$ C. $y = \frac{1}{2}x + 8$ D. $x + 2y = 8$	Work / thinking
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4B. Equivalent Forms | Version B

Which equation is equivalent to $y + 3 = 2(x - 1)$? A. $y = 2x + 1$ B. $y = 2x - 5$ C. $y = -2x - 5$ D. $2x + y = 5$	Work / thinking
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5A. Point-Slope Form | Version A

Using point-slope form, which equation has slope -2 and passes through (6, -4)? A. $y - 4 = -2(x - 6)$ B. $y + 4 = -2(x - 6)$ C. $y + 4 = -2(x + 6)$ D. $y - 4 = 1/2(x - 6)$	Work / thinking
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5B. Point-Slope Form | Version B

Using point-slope form, which equation has slope 3 and passes through (-2, 5)? A. $y - 5 = 3(x + 2)$ B. $y + 5 = 3(x - 2)$ C. $y - 2 = 3(x + 5)$ D. $y + 2 = 3(x - 5)$	Work / thinking
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6A. Equation Through Two Points | Version A

What is the slope-intercept equation through (10, 4) and (0, 2)? A. $y = 1/5x - 2$ B. $y = -1/5x + 6$ C. $y = 1/5x + 2$ D. $y = -5x + 54$	Work / thinking _____ _____ _____
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6B. Equation Through Two Points | Version B

What is the slope-intercept equation through (-5, 7) and (0, -3)? A. $y = -2x - 3$ B. $y = 2x - 3$ C. $y = -2x + 7$ D. $y = 1/2x - 3$	Work / thinking _____ _____ _____
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7A. Zero Slope | Version A

What type of line has a zero slope? A. Vertical line B. Horizontal line	Work / thinking _____ _____ _____
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7B. Undefined Slope | Version B

What type of line has an undefined slope? A. Vertical line B. Horizontal line	Work / thinking _____ _____ _____
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8A. Vertical and Horizontal Lines | Version A

The equation of a line is $x = 0$. Which describes this line? A. The line is vertical B. The line is horizontal	Work / thinking
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8B. Vertical and Horizontal Lines | Version B

The equation of a line is $y = -4$. Which describes this line? A. The line is vertical B. The line is horizontal	Work / thinking
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9A. Standard to Slope-Intercept | Version A

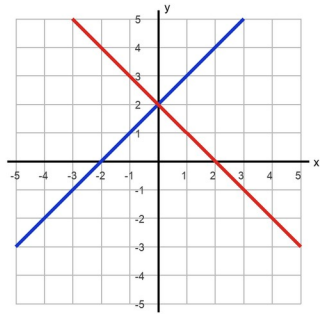
Convert $8x - 6y = 12$ to slope-intercept form.	Work / thinking
A. $y = -4/3x - 2$	_____
B. $y = 3/4x - 2$	_____
C. $y = -4/3x + 2$	_____
D. $y = 4/3x - 2$	

9B. Standard to Slope-Intercept | Version B

Convert $6x + 4y = 20$ to slope-intercept form.	Work / thinking
A. $y = -3/2x + 5$	_____
B. $y = 3/2x + 5$	_____
C. $y = -2/3x + 5$	_____
D. $y = -3/2x - 5$	

10A. Parallel or Perpendicular | Version A

The graphed lines have slopes 1 and -1. How are they related?

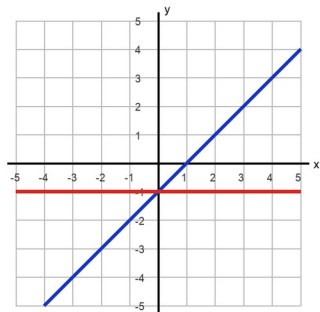


- A. Parallel
- B. Perpendicular
- C. Neither

Work / thinking

10B. Parallel or Perpendicular | Version B

The graphed lines have slopes 1 and 0. How are they related?



- A. Parallel
- B. Perpendicular
- C. Neither

Work / thinking

11A. Intersection of Two Lines | Version A

The graphs of $x = -3$ and $y = x - 3$ intersect at what point? A. (0, 0) B. (0, -3) C. (-3, 0) D. (-3, -3) E. (-3, -6)	Work / thinking
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11B. Intersection of Two Lines | Version B

The graphs of $x = 2$ and $y = -2x + 5$ intersect at what point? A. (2, 1) B. (1, 2) C. (2, -1) D. (-2, 1) E. (0, 5)	Work / thinking
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12A. Y-Intercept from Standard Form | Version A

What is the y-intercept of $7x + 2y = 14$?	Work / thinking
A. (0, 2)	_____
B. (7, 0)	_____
C. (0, 7)	_____
D. (2, 0)	

12B. Y-Intercept from Standard Form | Version B

What is the y-intercept of $5x - 4y = 20$?	Work / thinking
A. (0, 5)	_____
B. (0, -5)	_____
C. (4, 0)	_____
D. (-5, 0)	

13A. Testing Linear Inequalities | Version A

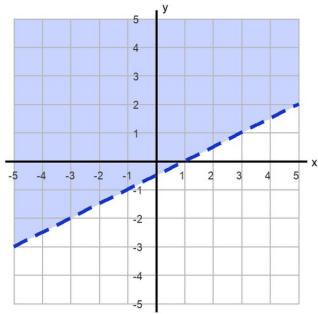
Which points satisfy $y > x + 3$? Select all that apply. A. (4, 0) B. (0, 1) C. (3, 7) D. (-4, 0) E. (-1, 3)	Work / thinking _____ _____ _____
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13B. Testing Linear Inequalities | Version B

Which points satisfy $y \leq -2x + 1$? Select all that apply. A. (0, 1) B. (1, 0) C. (-1, 3) D. (2, -3) E. (-2, 4)	Work / thinking _____ _____ _____
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14A. Inequality from a Graph | Version A

Which inequality is represented by the graph?

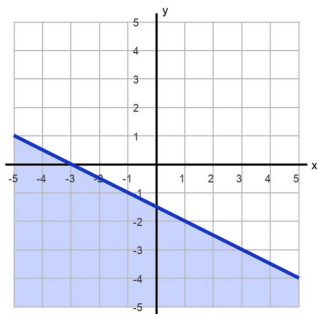


- A. $y > \frac{1}{2}x - \frac{1}{2}$
- B. $y < \frac{1}{2}x - \frac{1}{2}$
- C. $y \geq \frac{1}{2}x - \frac{1}{2}$
- D. $y \leq \frac{1}{2}x - \frac{1}{2}$

Work / thinking

14B. Inequality from a Graph | Version B

Which inequality is represented by the graph?



- A. $y < -\frac{1}{2}x - \frac{3}{2}$
- B. $y > -\frac{1}{2}x - \frac{3}{2}$
- C. $y \leq -\frac{1}{2}x - \frac{3}{2}$
- D. $y \geq -\frac{1}{2}x - \frac{3}{2}$

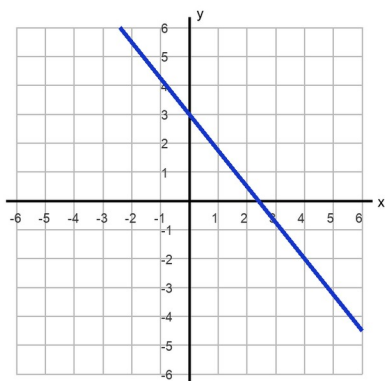
Work / thinking

Free-Response Practice

Show clear work. For graph problems, mark the slope, intercept, boundary line, or line of best fit you used.

15-A. Equation from a Graph in Standard Form | Version A

Write the equation of the graphed line in fully simplified standard form.

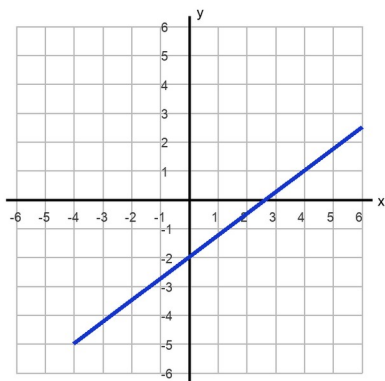


Answer: _____

Work / thinking

15-B. Equation from a Graph in Standard Form | Version B

Write the equation of the graphed line in fully simplified standard form.



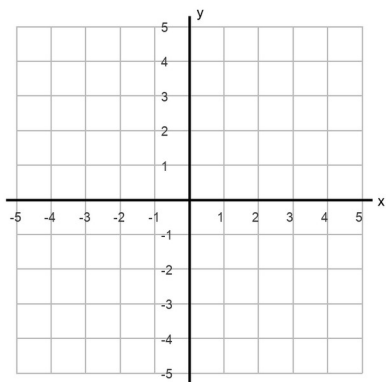
Answer: _____

Work / thinking

16-A. Graphing and Reading Function Notation | Version A

Graph $f(x) = -3x + 1$.

Then find: slope, y-intercept, $f(1)$, and x when $f(x) = 4$.



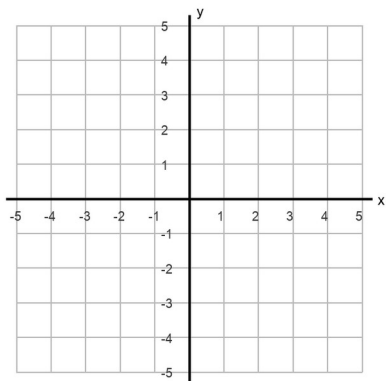
Answer: _____

Work / thinking

16-B. Graphing and Reading Function Notation | Version B

Graph $g(x) = 2x - 5$.

Then find: slope, y-intercept, $g(3)$, and x when $g(x) = 1$.



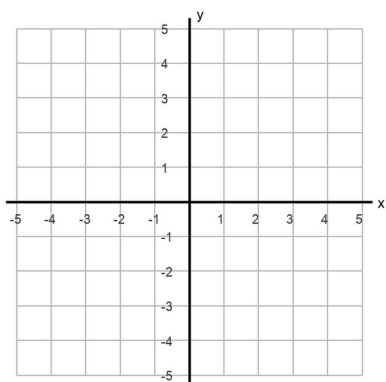
Answer: _____

Work / thinking

17-A. Graphing from Standard Form | Version A

Graph $2x + 3y = 6$ using intercepts.

Find the x-intercept, y-intercept, and equation in slope-intercept form.



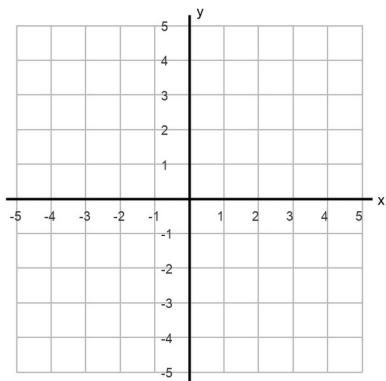
Answer: _____

Work / thinking

17-B. Graphing from Standard Form | Version B

Graph $4x - 2y = 8$ using intercepts.

Find the x-intercept, y-intercept, and equation in slope-intercept form.



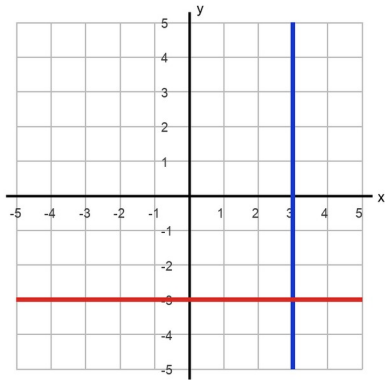
Answer: _____

Work / thinking

18-A. Vertical and Horizontal Lines | Version A

For each graphed line, state the slope and equation.

Blue line first; red line second.



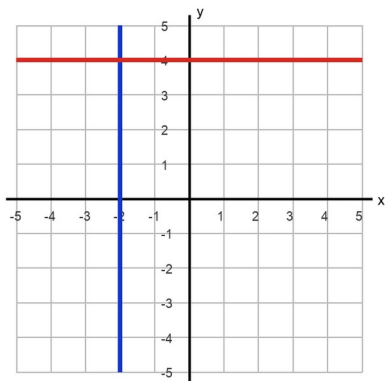
Answer: _____

Work / thinking

18-B. Vertical and Horizontal Lines | Version B

For each graphed line, state the slope and equation.

Blue line first; red line second.



Answer: _____

Work / thinking

19-A. Parallel Lines | Version A

Write the equation in slope-intercept form of the line through $(-2, 3)$ and parallel to $y = \frac{3}{2}x - 8$. Answer: _____	Work / thinking
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19-B. Parallel Lines | Version B

Write the equation in slope-intercept form of the line through $(4, -1)$ and parallel to $y = -\frac{1}{2}x + 7$. Answer: _____	Work / thinking
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20-A. Perpendicular Lines | Version A

Write the equation in slope-intercept form of the line through (9, -2) and perpendicular to $y = -\frac{3}{4}x + 5$. Answer: _____	Work / thinking
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20-B. Perpendicular Lines | Version B

Write the equation in slope-intercept form of the line through (-6, 5) and perpendicular to $y = 2x - 1$. Answer: _____	Work / thinking
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21-A. Matching Equations to Graphed Lines | Version A

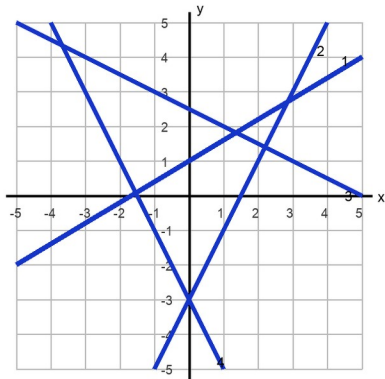
Four lines are graphed. Write the line number beside each equation.

$$y = \frac{3}{5}x + 1$$

$$y = 2x - 3$$

$$y = -\frac{1}{2}x + \frac{5}{2}$$

$$y = -2x - 3$$



Answer: _____

Work / thinking

21-B. Matching Equations to Graphed Lines | Version B

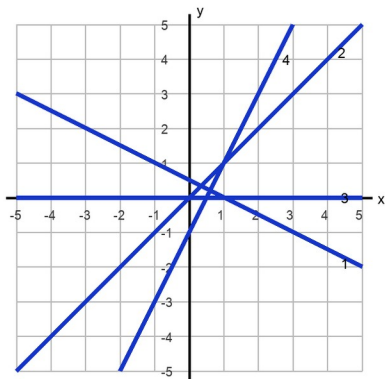
Four lines are graphed. Write the line number beside each equation.

$$y = -\frac{1}{2}x + \frac{1}{2}$$

$$y = x$$

$$y = 0$$

$$y = 2x - 1$$



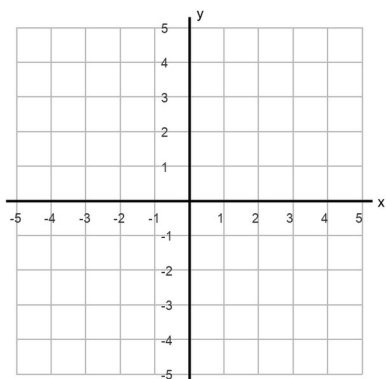
Answer: _____

Work / thinking

22-A. Graphing Linear Inequalities | Version A

Graph $y > -\frac{3}{4}x + 2$.

Use a test point to confirm the shading.



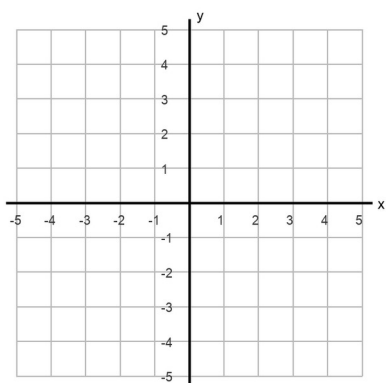
Answer: _____

Work / thinking

22-B. Graphing Linear Inequalities | Version B

Graph $y \leq \frac{2}{3}x - 1$.

Use a test point to confirm the shading.

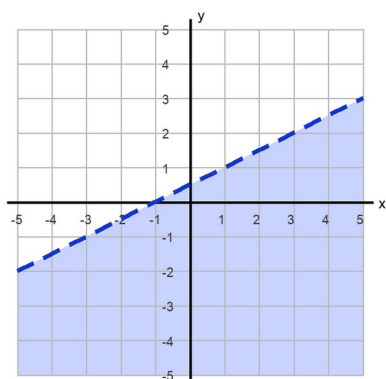


Answer: _____

Work / thinking

23-A. Writing Inequalities from Graphs | Version A

Write the inequality represented by the shaded graph.

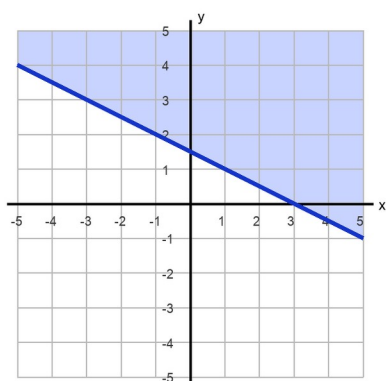


Answer: _____

Work / thinking

23-B. Writing Inequalities from Graphs | Version B

Write the inequality represented by the shaded graph.



Answer: _____

Work / thinking

Student Self-Check Answer Bank

Important: This is not a worked solution key. It is a self-check bank. If your answer is not here, go back and check the graph, boundary line, slope, intercept, or model setup.

Multiple-Choice Practice

Solve first, then check. Cross off each answer only after your work supports it.

B. The line is horizontal	A. -2
C. $y \leq -1/2x - 3/2$	A. Vertical line
B. $y + 4 = -2(x - 6)$	A. $y > 1/2x - 1/2$
B. (0, -5)	C. $y = 1/5x + 2$
C. (0, 7)	A. -5/2
D. $y = -2x + 4$	B. Horizontal line
A. $y = -1/2x + 8$	A. (2, 1)
C, D, E	B. $y = 2x - 5$
E. (-3, -6)	D. $y = 4/3x - 2$
A. $y - 5 = 3(x + 2)$	B. 3/4
A. $y = -2x - 3$	B. Perpendicular
B. -2	A, C, D, E
A. $y = -3/2x + 5$	A. $y = 2x - 3$
A. The line is vertical	C. Neither

Free-Response Practice

Solve first, then check. Cross off each answer only after your work supports it.

1, 2, 3, 4	$3x - 4y = 8$
x-intercept (2, 0); y-intercept (0, -4); $y = 2x - 4$	$y \geq -1/2x + 3/2$
$y = -1/2x + 2$	1, 2, 3, 4
$y = 3/2x + 6$	Solid boundary $y = 2/3x - 1$; shade below; (0, 0) is not a solution
x-intercept (3, 0); y-intercept (0, 2); $y = -2/3x + 2$	slope -3; y-intercept (0, 1); $f(1) = -2$; $x = -1$
blue: undefined slope, $x = -2$; red: slope 0, $y = 4$	$y = -1/2x + 1$
Dashed boundary $y = -3/4x + 2$; shade above; (0, 0) is not a solution	$5x + 4y = 12$
$y < 1/2x + 1/2$	slope 2; y-intercept (0, -5); $g(3) = 1$; $x = 3$
$y = 4/3x - 14$	blue: undefined slope, $x = 3$; red: slope 0, $y = -3$

Final Check Before the Assessment

- ☐ I can write and choose equations of lines from graphs.
- ☐ I can graph linear equations using slope and intercepts.
- ☐ I can write equations for parallel and perpendicular lines.
- ☐ I can graph and interpret linear inequalities.
- ☐ I can match equations to graphed lines and use function notation from a linear equation.